

Week 1 Content

Author.....	2
Introduction.....	3
Missing Values.....	4
Duplicate Records.....	5
Date Format Validation.....	6
Numeric Data Validation.....	7
Text Category Review.....	8
Final Deliverables.....	9
Data Cleaning Summary.....	10

Author

Senior Full-Stack Developer | Flutter Mobile Developer | Test Automation Engineer

Hello, my name is Julian. I am a Colombian Full-Stack and Mobile Application Developer with experience building scalable web and mobile solutions across a wide range of technologies. Throughout my career, I have successfully delivered more than 30 projects, working on both frontend and backend development, as well as test automation and quality assurance.

In addition to software development, I am a contemporary musician with over 15 years of experience in the music industry, which has strengthened my creativity, discipline, and ability to collaborate in diverse environments.

I am passionate about building high-quality software, creating meaningful user experiences, and continuously learning new technologies. I am currently open to new opportunities and always interested in connecting with professionals who share a passion for technology and innovation.

Contact Information

Email: contact@julianbelmonte.com

Phone: (+57) 312 562 9335

GitHub: <https://github.com/jucora>

Website: <https://julianbelmonte.com>

Introduction

Data quality is one of the most critical aspects of the data analytics lifecycle. Before performing any analysis, visualization, or predictive modeling, it is essential to ensure that the dataset is accurate, complete, and consistent. Poor data quality can lead to misleading insights, incorrect conclusions, and ineffective business decisions.

The objective of Project 1, Data Cleaning and Preparation, is to apply fundamental data cleaning techniques to a raw dataset and transform it into a reliable source of information. This process involves identifying and handling missing values, detecting duplicate records, validating data formats, and ensuring consistency across numerical and categorical fields.

For this project, a comprehensive review of the provided dataset was conducted to evaluate its overall quality and integrity. Several validation procedures were performed, including missing value analysis, duplicate detection, date format verification, numeric consistency checks, and categorical data standardization.

The outcome of this project is a cleaned and validated dataset that is suitable for future analytical tasks and decision making processes. The findings and actions taken during the cleaning process are documented in the following sections of this report.

Missing Values

A review of the dataset identified only one column containing missing values:

Column	Missing Values
CouponCode	309

All other columns contained **0 missing values**.

Implementation:

I replaced the blank entries with: **No Coupon**

This clearly indicates that the customer did not use a coupon during the transaction.

Duplicate Records

The dataset was examined for duplicate records and duplicate identifiers.

The validation results were:

Duplicate rows	Duplicate OrderIDs
0	0

Therefore, the dataset already satisfies the duplicate-record requirement.

- No duplicate records were found in the dataset.
- Duplicate **OrderID** count = 0.

Date Format Validation

The **Date** column was reviewed and found to be stored correctly as a date field.

No invalid or incorrectly formatted dates were identified.

To ensure consistency, the dates were standardized using the format:

YYYY-MM-DD

For example:

2025-05-08

This demonstrates that date formatting was reviewed and validated during the cleaning process.

Date
2023-01-04
2024-08-23
2024-02-27
2023-10-15
2025-05-08
2023-10-23
2025-06-17
2023-05-12
2025-04-02
2023-11-21
2023-12-29
2024-02-17
2024-10-15
2023-08-30
2023-03-27

Numeric Data Validation

A consistency check was performed on all records to verify the relationship between pricing fields.

The following rule was validated across all 1,200 rows:

TotalPrice = Quantity × UnitPrice

No inconsistencies were identified.

A	B	C	D	E	F	G	H	I	J	K	L	M	N
OrderID	Date	CustomerID	Product	Quantity	UnitPrice	ShippingAddress	PaymentMethod	OrderStatus	TrackingNumber	ItemsInCart	CouponCode	ReferralSource	TotalPrice
ORD2000	2023-01-04	C72649	Monitor	5	570.62	928 Main St	Debit Card	Shipped	TRK37947903	7	SAVE10	Instagram	2853.1
ORD2000	2024-08-23	C75739	Phone	2	151.35	823 Main St	Online	Shipped	TRK91186779	3	SAVE10	Referral	302.7
ORD2000	2024-02-27	C81728	Tablet	5	550.68	512 Main St	Credit Card	Cancelled	TRK42903982	8	FREESHIP	Email	2753.4
ORD2000	2023-10-15	C33540	Chair	1	273.19	275 Main St	Debit Card	Returned	TRK62788070	5	SAVE10	Facebook	273.19
ORD2000	2025-05-08	C81840	Printer	4	626.01	668 Main St	Online	Delivered	TRK29241424	8	SAVE10	Email	2504.04
ORD2000	2023-10-23	C37249	Phone	2	245.86	934 Main St	Credit Card	Shipped	TRK72976927	4	SAVE10	Instagram	491.72
ORD2000	2025-06-17	C83492	Laptop	1	664.42	986 Main St	Gift Card	Returned	TRK96417362	6	SAVE10	Facebook	664.42
ORD2000	2023-05-12	C41460	Monitor	5	149.55	706 Main St	Cash	Shipped	TRK78809193	9	FREESHIP	Facebook	747.75
ORD2000	2025-04-02	C26817	Phone	2	134.28	904 Main St	Gift Card	Cancelled	TRK61042692	2	No Coupon	Email	268.56
ORD2000	2023-11-21	C31946	Desk	4	509.38	102 Main St	Credit Card	Shipped	TRK33478363	6	SAVE10	Google	2037.52
ORD2000	2023-12-29	C43443	Tablet	5	625.97	333 Main St	Credit Card	Returned	TRK98859248	9	WINTER15	Instagram	3129.85
ORD2000	2024-02-17	C93861	Monitor	3	49.14	831 Main St	Online	Returned	TRK48234646	7	SAVE10	Email	147.42
ORD2000	2024-10-15	C38785	Monitor	2	180.5	179 Main St	Debit Card	Pending	TRK20419991	6	FREESHIP	Referral	361
ORD2000	2023-08-30	C88348	Laptop	3	201.49	903 Main St	Credit Card	Returned	TRK81913874	4	FREESHIP	Facebook	604.47
ORD2000	2023-03-27	C98474	Tablet	2	393.33	980 Main St	Debit Card	Pending	TRK79186539	3	SAVE10	Instagram	786.66
ORD2000	2023-07-17	C39416	Printer	1	473.96	942 Main St	Cash	Delivered	TRK54930938	3	No Coupon	Google	473.96
ORD2000	2023-01-24	C28976	Printer	2	533.81	914 Main St	Credit Card	Pending	TRK13194259	3	FREESHIP	Email	1067.62
ORD2000	2024-03-02	C60672	Tablet	1	423.4	382 Main St	Debit Card	Shipped	TRK11972116	3	No Coupon	Referral	423.4

Text Category Review

The categorical columns were reviewed for consistency and typographical errors.

PaymentMethod	• Cash • Credit Card • Debit Card • Gift Card • Online
OrderStatus	• Cancelled • Delivered • Pending • Returned • Shipped
ReferralSource	• Email • Facebook • Google • Instagram • Referral

No spelling mistakes or inconsistent variations were found, such as:

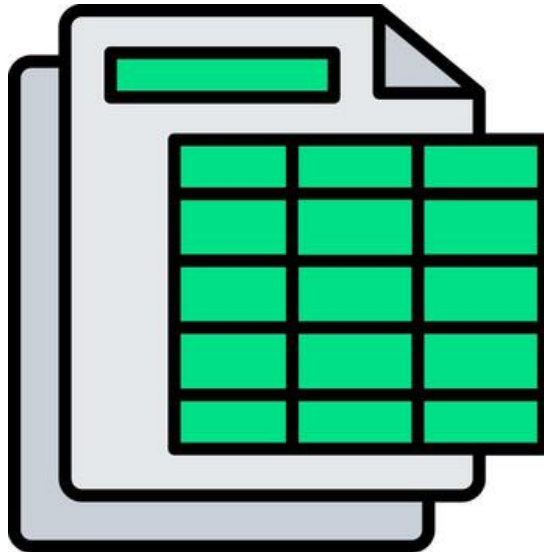
- Shipped
- shipped
- SHIPPED

All categorical values were already standardized.

Final Deliverables

The final cleaned dataset includes the following actions:

1. Replaced the 309 missing values in the **CouponCode** column with **No Coupon**.
2. Standardized the **Date** column using the **YYYY-MM-DD** format.
3. Performed a final duplicate check.
4. Saved the file as: [Cleaned_Dataset_Project1.xlsx](#)



Data Cleaning Summary

Task	Result
Missing Values	309 missing values found in CouponCode
Missing Values Handling	Replaced with "No Coupon"
Duplicate Records	0
Duplicate Order IDs	0
Invalid Dates	0
Numeric Validation	TotalPrice validated successfully
Final Rows	1200

The dataset appears to have been intentionally designed to test the identification and handling of missing values, particularly within the **CouponCode** column.

Duplicate records, date formats, and numerical calculations were already accurate and consistent.

The primary objective of this project was not to discover complex data quality issues, but rather to demonstrate a structured and professional data-cleaning process.

After addressing the missing values and validating the dataset, the data is considered clean, reliable, and ready for further analytical activities.